

PROJECT REPORT

Google Play Store Analytics Dashboard

using Microsoft Excel

~ PRAACHI AJMERA

Business Analytics Portfolio Project

Tool Used: Microsoft Excel (Pivot Tables, Pivot Charts, Slicers)

Dataset Source: Kaggle — Google Play Store Apps

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1. Project Overview

Overview

The Google Play Store hosts thousands of mobile applications across diverse categories such as Games, Education, Business, Health, Productivity, Finance, and Entertainment. With such a large volume of applications, it becomes difficult to manually identify trends related to user engagement, application quality, pricing strategies, and developer activity.

This project transforms raw Google Play Store application data into meaningful business insights using Microsoft Excel. The project covers the complete analytics lifecycle including data cleaning, feature engineering, exploratory data analysis (EDA), KPI development, and dashboard creation.

The final deliverable is an interactive Excel dashboard that enables users to filter applications dynamically and monitor important business metrics for better decision-making.

2. Business Problem

The Google Play Store contains a massive number of applications, but raw data alone provides little value for business decisions.

Developers, product managers, and business analysts often need answers to questions such as:

- Which application categories attract the highest installs?
- Which categories receive the highest user ratings?
- Are free applications more common than paid applications?
- How are applications distributed across different quality levels?
- How frequently are developers updating their applications?
- Which install ranges contain the majority of applications?

This project aims to convert raw application data into actionable business insights.

3. Project Objectives

The objectives of this project are to:

- Clean inconsistent and missing data.
- Transform raw data into an analysis-ready dataset.
- Create meaningful analytical features.
- Build business KPIs.
- Perform exploratory data analysis.
- Develop an interactive Excel dashboard.
- Generate business insights and recommendations.

4. Dataset Information

Dataset Name

Google Play Store Apps Dataset

Source

Kaggle

The dataset contains information about:

- Application Name
- Category
- Rating
- Reviews
- Size
- Installs
- Type
- Price
- Genres
- Last Updated
- Android Version

Total applications after cleaning: 10,356

5. Tools & Technologies

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Excel Functions
- Conditional Formatting
- Feature Engineering
- Dashboard Design

6. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Corrected the misaligned data row.
- Checked all columns for missing values.
- Converted Install values into numeric format.
- Converted Size values into MB.

- Standardized date fields.
- Verified column consistency.
- Preserved the original dataset and created a separate cleaned dataset.

7. Feature Engineering

The following analytical features were created:

Feature	Purpose
Installs_Clean	Numeric version of installs
Size_MB	App size converted into MB
Rating_Band	Categorized application ratings
Review_Band	Categorized review counts
Price_Clean	Numeric application price
Price_Band	Price segmentation
Update_Year	Extracted year from Last Updated

8. Dashboard KPIs

The dashboard summarizes six important performance indicators.

KPI	Value
Total Applications	10,356
Average Rating	4.19
Total Reviews	4,204 Million
Total Installs	146.6 Billion
Free Applications	9,591
Paid Applications	765

9. Dashboard Visualizations

The dashboard contains six interactive visualizations.

1. Top Categories by Total Installs

Purpose

Identifies application categories with the highest number of installs, helping understand user demand and market popularity.

2. Free vs Paid Applications

Purpose

Shows the distribution of free and paid applications, providing insight into pricing and monetization strategies.

3. Average Rating by Category

Purpose

Compares average user ratings across application categories to identify areas delivering better user satisfaction.

4. Apps by Rating Band

Purpose

Groups applications into rating categories such as Excellent, Good, Average, Poor, and No Rating. This helps evaluate the overall quality distribution of applications available on the Play Store.

5. Applications Updated by Year

Purpose

Shows yearly application update activity, helping identify developer maintenance trends within the dataset.

6. Applications by Install Band

Purpose

Groups applications into install ranges to understand how applications are distributed across different popularity levels.

10. Dashboard Filters

The dashboard includes interactive slicers that allow users to dynamically filter the analysis.

Available filters include:

- Category
- Type
- Rating Band

These filters automatically update all KPIs and charts, enabling quick exploration of different segments.

11. Key Business Insights

Insight 1

Game and Communication categories generate the highest number of installs, indicating strong user demand and high market engagement.

Insight 2

Approximately 93% of applications are free, while only 7% are paid. This indicates that developers predominantly rely on advertising, subscriptions, or in-app purchases rather than upfront payments.

Insight 3

The average application rating is 4.19, suggesting that most applications receive positive user feedback.

Insight 4

Most applications fall within the Good and Excellent rating bands, reflecting generally high application quality across the Play Store dataset.

Insight 5

Application updates increase significantly in the later years represented in the dataset, suggesting greater developer maintenance activity during that period. Since the dataset is historical, this trend reflects the available data rather than current Play Store activity.

Insight 6

Most applications belong to middle install bands such as 1M–10M and 100K–1M, while comparatively fewer applications achieve extremely high install counts. This highlights the competitive nature of the mobile application market.

12. Business Recommendations

Based on the dashboard findings, the following recommendations are suggested:

- Continue investing in high-demand categories such as Games and Communication while identifying opportunities in emerging categories with lower competition.
- Maintain high application quality to sustain strong user ratings and improve visibility in the Play Store.
- Optimize monetization strategies for free applications using advertisements, subscriptions, or in-app purchases.
- Encourage regular application updates to improve compatibility, security, and user engagement.
- Analyze applications within lower rating bands to identify common quality issues and prioritize improvements.
- Use the dashboard filters to compare performance across categories, application types, and rating bands before making product or marketing decisions.

13. Skills Demonstrated

This project demonstrates proficiency in:

- Data Cleaning
- Data Validation
- Feature Engineering
- Exploratory Data Analysis (EDA)
- KPI Development
- Pivot Tables
- Pivot Charts
- Dashboard Design
- Interactive Reporting
- Data Visualization
- Business Insight Generation
- Microsoft Excel

14. Project Limitations

- The dataset represents a historical snapshot and may not reflect the current state of the Google Play Store.
- Some missing values and inconsistent entries required preprocessing before analysis.
- User review sentiment analysis was outside the scope of this project.
- The dashboard focuses on descriptive analytics and does not include predictive modeling.

15. Future Enhancements

Future improvements could include:

- Recreate the dashboard using Power BI for enhanced interactivity.
- Integrate the Google Play Store user reviews dataset to perform sentiment analysis.
- Build predictive models to forecast download trends.
- Automate data preparation using Power Query.
- Add additional KPIs such as average price by category, estimated revenue potential, and review growth over time.

16. AI Usage Statement

AI Usage Disclosure

Generative AI (ChatGPT) was used as a learning assistant throughout this project. AI was used to explain Excel concepts, suggest data cleaning approaches, recommend analytical techniques, review dashboard layouts, and improve documentation. All data cleaning, feature engineering, Excel formulas, PivotTables, PivotCharts, dashboard implementation, validation, and final business interpretations were completed, verified, and customized by the project author.

17. Conclusion

This project demonstrates the end-to-end workflow of a Business Analyst using Microsoft Excel. Starting with raw Google Play Store data, the project applied data cleaning, feature engineering, exploratory analysis, KPI development, and interactive dashboard design to transform raw information into actionable business insights. The final dashboard enables users to analyze application performance through dynamic filters while demonstrating practical skills in data preparation, visualization, and business storytelling.

18. Business Impact

This dashboard provides a centralized view of key Google Play Store metrics, enabling faster analysis of application performance across categories, pricing models, quality levels, and popularity. By combining KPIs, interactive filters, and visualizations, stakeholders can quickly identify trends, compare market segments, and support evidence-based decisions related to product strategy, monetization, and application improvement. This demonstrates how descriptive analytics can transform raw data into actionable business intelligence.